THE COMPONENT MORPHISMS AS MORPHISMS OF RINGED TOPOI AND THE MODULE-LEVEL LEFT ADJOINTS G_{N!}

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the adjunction bijection (source lines 1503-1508) is obtained 'proceeding as in the proof of Lemma restriction-to-components-site' - that proof (delivered as card L46149) carries the corresponding construction in full, maps in both directions spelled out; the substitution of twisted copies f_varphi^*G for pullback sheaves and of semi-linearity for plain functoriality is the only change, and the transition construction is displayed in-span; the statement's caveat that g_{n!}G lands in modules with the given restriction formula is exactly what the displayed model proves.

Editorial note. The label lemma-restriction-module-to-components-site names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The component morphisms of topoi and their shrieks. Same chapter, \label{lemma-restriction-to-components-site} (card L46149 of this batch). Lines 1483-1486: g_n^{-1}O = O_n gives the ringed-topoi structure; the adjunction template of its proof is reused for the module bijection.
- The lower-shriek adjoint for ringed topoi. Modules on Sites chapter, \label{sites-modules-lemma-lower-shriek-modules}. Lines 1486-1488: existence of the left adjoint g_{n!} on modules.
- Sheaves of modules on the total site described componentwise. Same chapter, \label{lemma-describe-sheaves-simplicial-site-site}. Lines 1498-1501: the semi-linear transition maps define an O-module via the componentwise description.
- The ringed morphisms f_varphi. Same chapter, the morphism of ringed topoi f_varphi using O(varphi) : f_varphi^{-1}O_n -> O_m. Lines 1470-1476: the twist f_varphi^* used in the formula for the restriction of g_{n!}G.

Source reference. Original reference 09WF: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 953–953. Prerequisite pointer invoked at proof

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line(s) 1507; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09WG: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 970–970. Prerequisite pointer invoked at proof line(s) 1466; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0797: sources/stacks-chow/tags.txt, source lines 9262–9262. Prerequisite pointer invoked at proof line(s) 1487; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09WE: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 888–888. Prerequisite pointer invoked at proof line(s) 1460; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 09WE. Let $\mathcal{O}$ be a sheaf of rings on $\mathcal{C}_{total}$. There is a canonical morphism of ringed topoi $g_n : (Sh(\mathcal{C}_n), \mathcal{O}_n) \to (Sh(\mathcal{C}_{total}), \mathcal{O})$ agreeing with the morphism g_n of Lemma 09WG on underlying topoi. The functor $g_n^* : Mod(\mathcal{O}) \to Mod(\mathcal{O}_n)$ has a left adjoint $g_{n!}$. For $\mathcal{G}$ in $Mod(\mathcal{O}_n)$-modules the restriction of $g_{n!}\mathcal{G}$ to $\mathcal{C}_m$ is*

$$\bigoplus\nolimits_{\varphi:[n]\to[m]} f_\varphi^*\mathcal{G}$$

where $f_\varphi : (Sh(\mathcal{C}_m), \mathcal{O}_m) \to (Sh(\mathcal{C}_n), \mathcal{O}_n)$ is the morphism of ringed topoi agreeing with the previously defined f_φ on topoi and using the map $\mathcal{O}(\varphi) : f_\varphi^{-1}\mathcal{O}_n \to \mathcal{O}_m$ on sheaves of rings.

Proof. By Lemma 09WG we have $g_n^{-1}\mathcal{O} = \mathcal{O}_n$ and hence we obtain our morphism of ringed topoi. By Modules on Sites, Lemma 0797 we obtain the adjoint $g_{n!}$. To prove the formula for $g_{n!}$ we first define a sheaf of $\mathcal{O}$-modules $\mathcal{H}$ on $\mathcal{C}_{total}$ with degree m component the $\mathcal{O}_m$-module

$$\mathcal{H}_m = \bigoplus\nolimits_{\varphi:[n]\to[m]} f_\varphi^*\mathcal{G}$$

Given a map $\psi : [m] \to [m']$ the map $\mathcal{H}(\psi) : f_\psi^{-1}\mathcal{H}_m \to \mathcal{H}_{m'}$ is given on components by

$$f_\psi^{-1}f_\varphi^*\mathcal{G} \to f_\psi^*f_\varphi^*\mathcal{G} \to f_{\psi\circ\varphi}^*\mathcal{G}$$

Since this map $f_\psi^{-1}\mathcal{H}_m \to \mathcal{H}_{m'}$ is $\mathcal{O}(\psi) : f_\psi^{-1}\mathcal{O}_m \to \mathcal{O}_{m'}$-semi-linear, this indeed does define an $\mathcal{O}$-module (use Lemma 09WF). Then one proves directly that

$$\mathrm{Mor}_{\mathcal{O}_n}(\mathcal{G}, \mathcal{F}_n) = \mathrm{Mor}_{\mathcal{O}}(\mathcal{H}, \mathcal{F})$$

proceeding as in the proof of Lemma 09WG. Thus $\mathcal{H} = g_{n!}\mathcal{G}$ as desired. □